【物理化学特論】4. Vibrational and rotational energy

No. : ________ Name : ______________________________

(1) Draw the energy levels in a diagram for vibrational and rotational motions.

(2) Calculate the lowest excitation energies [J] of the vibrational and rotational motions of HCl molecule. The mass numbers of H and Cl atoms are 1.0 and 35.5, the bond length and the force constant of HCl molecule is 127 pm and 5.2×10^2 N/m, respectively.

(3) Calculate the wavelengths of photons absorbed at the vibrational and rotational excitations in (2).

(4) Determine the probability distribution functions $P_v(x)$ for a harmonic oscillator (mass m and spring constant k) for v=0 and 1. Draw the graphs of $P_v(x)$ for v=0 and 1.

(5) Prove that the mean deviation is zero ($\langle x \rangle$ =0) for harmonic oscillator for v=0.